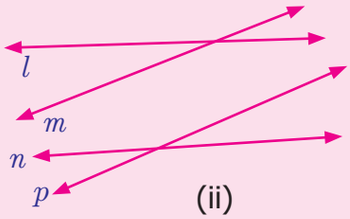
Try these

1. Draw as many possible transversals in the given figures.



(i)



(ii)

2. Draw a line which is not the transversal to the above figures.
3. How many transversals can you draw for the following two lines?



5.3.1. Angles formed by a transversal

If a transversal meet two lines, eight angles are formed at the points of intersection as shown in the Fig.5.15.

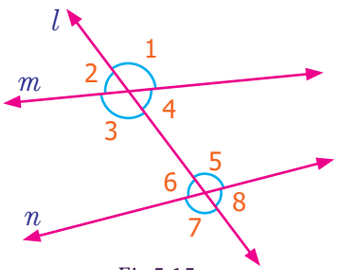


*Fig.*5.15

It is clear that the pairs of angles $\angle 1$, $\angle 2$, ; $\angle 3$, $\angle 4$, ; $\angle 5$, $\angle 6$ and $\angle 7$, $\angle 8$ are linear pairs. Can you find more linear pairs of angles?

Besides, the pairs $\angle 1$, $\angle 3$; $\angle 2$, $\angle 4$; $\angle 5$, $\angle 7$ and $\angle 6$, $\angle 8$ are vertically opposite angles.

We can further classify the angles shown in the Fig. 5.15 into different categories as follows.

Corresponding angles

Observe that the pair of angles $\angle 1$ and $\angle 5$ that are marked at the right side of the transversal l. In that $\angle 1$ lies above the line m and $\angle 5$ lies above the line n.

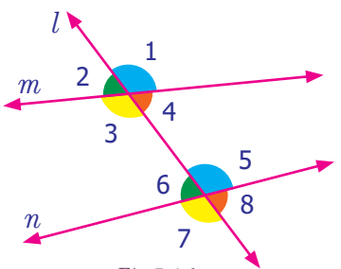


*Fig.*5.16

Also observe the pair of angles $\angle 2$ and $\angle 6$ that are marked on the left of the transversal l. In that $\angle 2$ lies above m and $\angle 6$ lies above n.

In the same way observe the pair of angles $\angle 3$ and $\angle 7$ that are marked on left of transversal l. In that $\angle 3$ lies below m and $\angle 7$ lies below n.

Observe the pair of angles $\angle 4$ and $\angle 8$ that are marked on the right of transversal l. In that $\angle 4$ lies below m and $\angle 8$ lies below n.

So all these pairs of angles have different vertices, lie on the same side (left or right) of the transversal(l), lie above or below the lines m and n. Such pairs are called corresponding angles.

7th_Maths_T1_EM_Chp5.indd 94 10/12/2022 18:13:39